

Intermittent Control of Multi-Agent Systems with Delayed Impulses: Application to Quadrotor UAVs

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Abstract—This study investigates the leader-following consensus of multi-agent systems via aperiodic intermittent control with delayed impulses. Based on the concept of average dwell time (ADT) for switched systems, the concept of average control width (ACW) in limiting form is proposed for intermittent control systems. Then, by using concepts of both ADT and ACW, we present some flexible sufficient conditions for leader-following consensus of multi-agent systems via average-width intermittent control. In particular, the delayed impulses at “on-off” switching instants on the intermittent control, which were not well addressed in the previous results, are studied in this paper. It is shown that under some conditions, the delay in impulses at “off” switching instants may result in consensus-disruptive impact on the addressed multi-agent systems. More interestingly, it is verified that the delay in impulses at “on” switching instants can have consensus-facilitating impact, namely, an originally inconsistent multi-agent system can achieve consensus by increasing this delay. Subsequently, the theoretical results are applied to the quadrotor unmanned aerial vehicles (UAVs). Finally, the effectiveness of the derived results is verified with two illustrative examples.

Index Terms—Leader-following consensus; Multi-agent system; Aperiodic intermittent control; Average dwell time; Average control width; Delayed impulse.

I. INTRODUCTION

Multi-agent systems have garnered significant attention in both theoretical and practical domains due to their diverse applications in areas such as robotics, sensor networks, and distributed computing. Over the past few decades, consensus control for multi-agent systems has been extensively studied by numerous researchers [1–5]. This interest is largely driven by applications in coordination, formation control, and resource allocation. Actually, the issue of leader-following consensus control itself is full of significance. Up to now, many

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notable control approaches have been studied for the leader-following consensus of multi-agent systems, e.g., impulsive control [2], intermittent control [3], event-triggered control [4], and so forth. Intermittent control approach proposed in [6] has been extensively investigated and applied to many systems (see [7–10]). Unlike continuous control requiring constant actuation, these systems typically employ intermittent controllers, which activate only when necessary. Based on the variability of control durations and intervals, intermittent strategies can be classified into two types: periodic and aperiodic intermittent control (AIC).

Compared to periodic intermittent control, AIC offers greater flexibility and practical applicability in certain scenarios, which motivates its adoption in this study. On the other hand, the consensus and stabilization issue via AIC has attracted interesting attention in recent years. For example, an innovative AIC in [11] employs adaptive dynamic programming to achieve optimal consensus in multi-agent systems. Meanwhile, [12] designed an AIC scheme aimed at fixed-time consensus in nonlinear multi-agent systems. Although many interesting results on AIC for multi-agent systems consensus have been reported, there are still two main aspects to be noted. On the one hand, traditional methods require that each “work” (control-on) duration is roughly similar (bounded by upper and lower limits). In contrast, the proposed average control width (ACW) approach is only concerned with the average duration of each “work” period over a sufficiently long time horizon. Even if some “work” periods are exceptionally short while others are exceptionally long, the system can maintain the desired performance as long as their mean value meets the required condition. This enable the controller greater flexibility in scheduling its “work” periods (see Section II for detailed discussions). Although the average control rate concept recently proposed by [13] provides a global characterization of control width distributions in AIC, ACW can be further refined into a more intuitive and easily verifiable condition.

On the other hand, the “on-off” switching behavior in intermittent control for the systems consensus may bring the impulses, which are usually described by abrupt state jumps (see [14–22]) at switching instants [23]. It is worth noting that there are clear differences between the impulsive control studied in this paper and the impulsive maneuver control in the aerospace field (e.g., orbital attack-defense game scenarios [24]). The former focuses on the consensus control of multi-agent systems, with the core being to achieve cooperative convergence in time-delay environments through aperiodic impulses adapted to the ACW framework; the latter targets spacecraft orbital maneuvers, aiming at the precise

adjustment of orbital parameters, which must strictly follow orbital mechanics laws and is subject to strong engineering constraints such as fuel and thrust limitations [24]. Although some interesting results on intermittent control systems with impulses have been reported (see [25–27]), there exist some constraints of upper bound on either control widths or impulsive intervals. Obviously, such constraints may lead to the inflexibility of the corresponding consensus criteria in these previous results.

Time delays are common phenomenon in dynamic systems, and in UAV systems mainly arise from wireless communication link jitter, formation interaction message forwarding, and onboard command response latency, which are thus characterized by randomness and short-term fluctuations. In contrast, time delays in the spacecraft domain are dominated by ground-space communication latency and attitude control command latency, with the typical features of long duration and high stability [5, 28, 29]. Beyond these domain-specific time delay characteristics, impulsive signals inherently induce time delays during their sampling, transmission, and generation processes [30]. For this reason, the effects of delayed impulses on the consensus and stability of various dynamical systems have been extensively investigated in the past decades [15, 31–35]. To the best of our knowledge, the delay effect on impulses has not been considered in the previous results (e.g. [25–27]) on intermittent control systems. It should be noted that if ignoring the effects of delayed impulses on intermittent control systems, we may derive some false conclusions. A common practice in existing research on delayed or impulsive systems is to model delays from different sources as a single aggregate term, τ , or to analyze only their average effect [15, 31–35]. However, this approach may obscure a key fact: delays with different physical mechanisms (e.g., “on” and “off” switching) and occurring at different time instants can have distinctly different effects on system performances. One of the motivations of this work is to decouple and finely characterize the differential impacts of these two types of delays (detailed discussions can be seen in Section V). This study further investigates the impact of impulsive delays on system performance under intermittent control.

Motivated by the aforementioned facts, the concept of ACW in limiting form, where the restrictions on the lower/upper bound of control widths can be relaxed, is proposed to characterize the control widths in all control periods from a holistic perspective. Compared with the concept of average control rate, it is shown that the concept of ACW is more intuitive and easier to check in some situations. Based on the concepts of ADT and ACW in limiting form, some flexible leader-following consensus criteria are established for multi-agent systems under AIC, which can be effectively applied to quadrotor unmanned aerial vehicles (UAVs) [36–38]. More importantly, the effects of delayed impulses on consensus of the considered multi-agent systems under intermittent control are fully revealed. The main contributions of the current study are listed as follows:

- (1) In order to globally address the control widths in all control periods, the concept of ACW is proposed for AIC.

Diverging from average control rate that describes the global distribution pattern of control intervals, the ACW characterizes the mean size of each activation period. This alternative concept is arguably more intuitive and easier to verify in some sense.

- (2) Based on the concepts of ADT and ACW in limiting form, the restrictions on lower/upper bounds of both control periods and control widths are relaxed and the average-width intermittent control is introduced. Further, some flexible Lyapunov-based conditions are established to ensure leader-following consensus in multi-agent systems under the introduced average-width intermittent control framework. The proposed control strategy is then experimentally validated through quadrotor UAVs implementations, demonstrating both theoretical soundness and practical efficacy.
- (3) The current study investigates the effects of delayed impulses that result from “on-off” switching behaviors on intermittent consensus control. It is shown that the delay in impulses at “off” switching instants may have a negative impact on the multi-agent systems under intermittent control. Surprisingly, it is also shown that the delay in impulses at “on” switching instants can bring originally inconsistent multi-agent systems into consensus.

The remainder of the study is arranged as follows. Section II introduces the leader-following consensus issue, intermittent control and necessary notations as well as definitions. The main results are introduced in Section III. An application to the quadrotor UAVs is presented in Section IV. Some numerical simulations and summaries are presented at last.

II. PRELIMINARIES

This paper adopts the following notation: $\mathbb{R}$ (real numbers), $\mathbb{R}_+$ (nonnegative real numbers), and $\mathbb{Z}_+$ (nonnegative integers). Furthermore, $\mathbb{R}^n$ and $\mathbb{R}^{n \times m}$ represent the spaces of n -dimensional real vectors and $n \times m$ real matrices, respectively. In particular, assume that matrices possess appropriate dimensions in this paper if not especially stated. Denote by the notation $\|\cdot\|$ the Euclidean norm of vectors. Moreover, $\mathbf{I}_n$ stands for the n -dimensional identity matrix. Let $\mathbf{P}^T$ and $\mathbf{P}^{-1}$ represent the transpose and the inverse of the square matrix $\mathbf{P}$, respectively. The symmetric block in a symmetric matrix is represented by $*$ and $|\cdot|$ stands for the absolute value function. The operator $\otimes$ is adopted to characterize the Kronecker product operation between two matrices.

This study considers the leader-following consensus issue of the following dynamical equation:

$$\dot{\mathbf{x}}_i(t) = \mathbf{A}\mathbf{x}_i(t) + \mathbf{B}\mathbf{u}_i(t), \quad i = 1, \dots, N, \quad t \geq t_0 \quad (1)$$

and

$$\dot{\mathbf{x}}_0(t) = \mathbf{A}\mathbf{x}_0(t), \quad t \geq t_0 \geq 0, \quad (2)$$

where the state vectors $\mathbf{x}_i$ (followers) and $\mathbf{x}_0$ (leader) belong to $\mathbb{R}^n$, and the control input $\mathbf{u}_i \in \mathbb{R}^m$.

The interaction topology of a leader-following multi-agent system (leader 0, followers 1 to N) is modeled by a digraph $\mathcal{G}_1 = (V_1, \mathcal{E}, \mathcal{A})$, with $V_1 = \{0, 1, \dots, N\}$. The adjacency

matrix $\mathcal{A} = [\bar{a}_{ij}]$ defines connection weights ($\bar{a}_{ij} > 0$ if and only if agent i receives data from j). The follower Laplacian $\mathbf{L} \in \mathbb{R}^{N \times N}$ has diagonal entries $l_{ii} = \sum_{j=1}^N \bar{a}_{ij}$ and off-diagonals $l_{ij} = -\bar{a}_{ij}$ ($i \neq j$). The leader's influence is captured by a diagonal matrix $\mathbf{C} = \text{diag}(\bar{a}_{10}, \dots, \bar{a}_{N0})$. Accordingly, the matrix $\mathbf{H}$ is defined as $\mathbf{L} + \mathbf{C}$.

In general, (1) and (2) are said to achieve leader-following consensus provided that

$$\lim_{t \rightarrow +\infty} \|\mathbf{x}_i(t) - \mathbf{x}_0(t)\| = 0, \quad i = 1, \dots, N,$$

where $\mathbf{x}_i(t)$ and $\mathbf{x}_0(t)$ stand for the solutions to (1) and (2), respectively.

For convenience, some assumptions are introduced before establishing the leader-following consensus criteria.

Assumption 1 ([39]). *In the digraph $\mathcal{G}_1$, node 0 can reach every node $i = 1, \dots, N$.*

Assumption 2. *The pair $(\mathbf{A}, \mathbf{B})$ is stabilizable.*

Remark 1. Assumption 1 permits digraph $\mathcal{G}_1$ to be either directed or undirected. Consequently, for the M-matrix $\mathbf{H}$, it is possible to construct a diagonal matrix $\mathbf{D} = \text{diag}(d_1, \dots, d_N)$ with $d_i > 0, i = 1, \dots, N$ such that $\mathbf{DH} + \mathbf{H}^T \mathbf{D}$ is positive definite [40].

Given that the pair $(\mathbf{A}, \mathbf{B})$ is stabilizable and $(\mathbf{I}_n, \mathbf{A})$ is observable, a unique positive definite solution $\mathbf{P}$ exists for the Riccati equation presented below [41]:

$$\mathbf{P}\mathbf{A} + \mathbf{A}^T \mathbf{P} - \rho_1 \mathbf{P}\mathbf{B}\mathbf{B}^T \mathbf{P} + \rho_2 \mathbf{I}_n = 0 \quad (3)$$

where ρ_1 and ρ_2 are positive real numbers.

Denote consensus error by $\mathbf{e}_i(t) = \mathbf{x}_i(t) - \mathbf{x}_0(t)$, and the AIC $\mathbf{u}_i(t)$ is devised as follows:

$$\begin{cases} \mathbf{u}_i(t) = \mathbf{K} \sum_{j=0}^N \bar{a}_{ij} (\mathbf{x}_j(t) - \mathbf{x}_i(t)), & t \in [t_k, t_k + \varsigma_k), \\ \mathbf{u}_i(t) = 0, & t \in [t_k + \varsigma_k, t_{k+1}), \end{cases} \quad (4)$$

where $\mathbf{K} \in \mathbb{R}^{m \times n}$ denotes the control gain matrix. Referring to the previous results on intermittent control with impulses [25–27] and considering the delay effects on such impulses, we investigate the leader-following consensus issue of multi-agent systems under intermittent control with delayed impulses at “on-off” switching instants $t_k + \varsigma_k$ and t_{k+1} , where $k \in \mathbb{Z}_+$:

$$\begin{cases} \mathbf{x}_i(t_k + \varsigma_k) = \mathbf{H}_k \mathbf{x}_i((t_k + \varsigma_k - \tau_{off})^-) \\ \quad - \mathbf{H}_k \mathbf{x}_0((t_k + \varsigma_k - \tau_{off})^-) + \mathbf{x}_0(t_k + \varsigma_k), \\ \mathbf{x}_i(t_{k+1}) = \mathbf{G}_k \mathbf{x}_i((t_{k+1} - \tau_{on})^-) \\ \quad - \mathbf{G}_k \mathbf{x}_0((t_{k+1} - \tau_{on})^-) + \mathbf{x}_0(t_{k+1}). \end{cases} \quad (5)$$

Remark 2. In intermittent control framework for multi-agent UAV systems, impulsive state jumps are an intrinsic consequence of the controller's mode switching behavior between the “on” and “off” states at the predefined switching instants [14–16]. In practical UAV systems, the controller cannot access the precise system state instantaneously at the planned switching instant t_{k+1} . Actually, the acquisition of state information involves sensor sampling, data conversion, and often

network transmission, all of which introduce inevitable delays (see [30, 31]). Therefore, the control command computed and applied at time t_{k+1} (which causes the impulsive state jump) is effectively based on the historical system state at time $t_{k+1} - \tau$.

The matrices $\mathbf{H}_k$ and $\mathbf{G}_k$ can be designed as deterministic control parameters. For instance, setting $\mathbf{H}_k = \mathbf{G}_k = \mathbf{I}$ corresponds to ideal switching behavior, while $\|\mathbf{H}_k\| < 1$ is adopted to construct a stabilizing impulse for error attenuation (see [42]). Actually, these matrices may also be equivalently derived from the inherent hardware or communication characteristics of the system. In such cases, $\mathbf{H}_k$ and $\mathbf{G}_k$ are determined by the physical properties of the actual system, and $\|\mathbf{H}_k\| > 1$ corresponds to a destabilizing impulse.

Further, the corresponding dynamics of consensus error $\mathbf{e}(t)$ can be described as follows:

$$\begin{cases} \dot{\mathbf{e}}(t) = [(\mathbf{I}_N \otimes \mathbf{A}) - (\mathbf{H} \otimes \mathbf{BK})]\mathbf{e}(t), & t \in (t_k, t_k + \varsigma_k), \\ \mathbf{e}(t_k + \varsigma_k) = (\mathbf{I}_N \otimes \mathbf{H}_k)\mathbf{e}((t_k + \varsigma_k - \tau_{off})^-), \\ \dot{\mathbf{e}}(t) = (\mathbf{I}_N \otimes \mathbf{A})\mathbf{e}(t), & t \in (t_k + \varsigma_k, t_{k+1}), \\ \mathbf{e}(t_{k+1}) = (\mathbf{I}_N \otimes \mathbf{G}_k)\mathbf{e}((t_{k+1} - \tau_{on})^-), \end{cases} \quad (6)$$

for all $k \in \mathbb{Z}_+$, where $\mathbf{e}(t) = \text{col}(\mathbf{e}_1(t), \dots, \mathbf{e}_N(t))$.

Clearly, error system (6) can be viewed as a switched system that has two subsystems subject to delayed impulses. In this study, the solutions to error system (6) are supposed to be right-continuous, that is, $\mathbf{e}(t_k^+) = \mathbf{e}(t_k)$ and $\mathbf{e}((t_k + \varsigma_k)^+) = \mathbf{e}(t_k + \varsigma_k)$ for all $k \in \mathbb{Z}_+$. To avoid the Zeno phenomenon, the intermittent control sequence $\{t_k, \varsigma_k\}_{k \in \mathbb{Z}_+}$ is assumed to meet that $0 \leq t_0 < t_0 + \varsigma_0 < t_1 < t_1 + \varsigma_1 < t_2 < \dots < t_k + \varsigma_k \rightarrow +\infty$ as $k \rightarrow +\infty$. In general, $t_{k+1} - t_k$ and ς_k are said to be the k -th control period and the k -th control width, respectively. In addition, the delays τ_{off} and τ_{on} are assumed to satisfy $\tau_{off} < \varsigma_k$ and $\tau_{on} < t_{k+1} - t_k - \varsigma_k$.

It is known that the concept of ADT introduced in [43] for characterizing the switching frequency of switched systems has been widely studied in the past two decades (see [17, 44, 45]). In particular, the ADT concept in limiting form was proposed in [46] for the more general switched systems. This concept is presented in the following for intermittent control systems.

Definition 1 ([46]). *The average dwell time of intermittent control sequence $\{t_k, \varsigma_k\}_{k \in \mathbb{Z}_+}$ is defined as follows:*

$$T_a = \lim_{t \rightarrow +\infty} \frac{t - t_0}{N(t, t_0)}, \quad (7)$$

where $N(t, t_0)$ represents the number of control periods of intermittent control on the interval $(t_0, t]$.

We can observe that condition (7) is more general than previous condition in the form of inequality, i.e., $\frac{t - t_0}{T_a} - N_0 \leq N(t, t_0) \leq \frac{t - t_0}{T_a} + N_0$, where $N_0 > 0$ stands for the elastic coefficient. Clearly, from the condition in the form of inequality, we can conclude that $T_a = \lim_{t \rightarrow +\infty} \frac{t - t_0}{N(t, t_0)}$ (see [46]), which yields (7). Note that some specific switching instant sequences (e.g., $N(t, t_0) = \ln(t - t_0)$) can be investigated by condition (7) but not by condition in the form of inequality (some more details can be seen in [46]).

Inspired by the concept of ADT in limiting form, the concept of ACW in limiting form is proposed to characterize the control widths from a holistic perspective.

Definition 2. Assuming the limit exists, the average control width of intermittent control sequence $\{t_k, \varsigma_k\}_{k \in \mathbb{Z}_+}$ is defined as follows:

$$\bar{\varsigma} = \lim_{t \rightarrow +\infty} \frac{\varsigma_0 + \varsigma_1 + \cdots + \varsigma_{N(t, t_0)-1}}{N(t, t_0)}, \quad (8)$$

where $N(t, t_0)$ is same as that defined in Definition 1.

Remark 3. The finite-time formulation of (8) can be given by $\bar{\varsigma}N(t, t_0) - \tau_0 \leq \varsigma_0 + \varsigma_1 + \cdots + \varsigma_{N(t, t_0)-1} \leq \bar{\varsigma}N(t, t_0) + \tau_0$, where $\tau_0 > 0$ is termed the chatter bound, and the remaining parameters are identical to those in (8). It is evident that condition (8) (in its limiting form) is less conservative. Moreover, the limiting expression $\bar{\varsigma} = \lim_{t \rightarrow +\infty} \frac{\varsigma_0 + \varsigma_1 + \cdots + \varsigma_{N(t, t_0)-1}}{N(t, t_0)}$ can be derived from this finite-time inequality.

Remark 4. In the most of previous results, control period $t_{k+1} - t_k$ and control width ς_k , $k \in \mathbb{Z}_+$ at least one of them is assumed to assign with a strict lower/upper bound. Recently, a concept of average control rate was introduced in [13] to relax the restrictions on the lower/upper bound of control widths. To be specific, AIC (4) is said to be with average control rate $\vartheta \in (0, 1)$, provided that there exists a scalar $N_0 > 0$ satisfying $N(t, s) \geq \vartheta(t - s) - N_0$, $\forall t > s \geq t_0$, where $N(t, s)$ stands for the sum of control widths in $(s, t]$. Technically speaking, compared with the concept average control rate, the concept of ACW is more intuitive and easier to verify. A demonstrative example shall be introduced in Section V.

Motivated Example: Consider a AIC sequence $\{t_k, \varsigma_k\}_{k \in \mathbb{Z}_+}$ defined by $t_k = 5k + 0.1 \times (-1)^k$ and $\varsigma_k = 3 + 1.6 \times (-1)^k$.

Following Definition 1 and Definition 2, ADT can be calculated as $T_a = 5$ and ACW $\bar{\varsigma} = 3$. However, the upper bound of ς_k is 4.6 and the lower bound is 1.4, and it is evident that the use of either bound as a reference will lead to a conservative result.

The information of the average control rate is not as intuitive as that of the average control width (ACW). Actually, it is insufficient to only use the information of average control rate when the effect of delayed impulses is considered. The derivation of the corresponding consensus criterion requires the integration of the specific information of the ACW, with detailed steps provided in Section III.

In this study, controller (4) that satisfies both (7) and (8) is said to be the average-width intermittent controller $(K, \{t_k\}, \{\varsigma_k\})$ with ADT constant T_a and ACW constant $\bar{\varsigma}$.

III. MAIN RESULTS

In the current section, criteria shall be established for the leader-following consensus of multi-agent systems (1) and (2) with delayed impulses (5) via AIC (4). Especially, by utilizing the concepts of ADT and ACW in limiting form, some flexible conditions on both the control periods and control widths shall be derived. Further, the effects of delayed impulses on the consensus of the considered multi-agent systems are fully revealed. Before giving our main result, we introduce

some notations. Let $d_M = \lambda_{\max}(\mathbf{D})$, $\lambda_1 = \lambda_{\min}(\mathbf{H}^T \mathbf{D} + \mathbf{D}\mathbf{H})$, $\gamma = \frac{\rho_1 d_M}{\lambda_1}$ and $\alpha_1 = \frac{\rho_2}{\lambda_{\max}(\mathbf{P})}$.

Theorem 1. Under Assumptions 1-2, suppose that given constants $\alpha_2 > 0$, $\beta_1 > 0$ and $\beta_2 > 0$, the following LMIs hold:

$$\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} - \alpha_2 \mathbf{P} < 0 \quad (9)$$

$$\begin{pmatrix} -\beta_1 \mathbf{P} & \mathbf{H}_k^T \mathbf{P} \\ * & -\mathbf{P} \end{pmatrix} < 0, \quad \begin{pmatrix} -\beta_2 \mathbf{P} & \mathbf{G}_k^T \mathbf{P} \\ * & -\mathbf{P} \end{pmatrix} < 0, \quad (10)$$

and

$$\ln \beta_1 + \ln \beta_2 + \alpha_1(\tau_{off} - \bar{\varsigma}) + \alpha_2(T_a - \bar{\varsigma} - \tau_{on}) < 0. \quad (11)$$

Then, leader-following consensus of systems (1) and (2) with delayed impulses (5) can be achieved via average-width intermittent controller $(\mathbf{K}, \{t_k\}, \{\varsigma_k\})$, where $\mathbf{K} = \gamma \mathbf{B}^T \mathbf{P}$.

Proof. Lyapunov function is used to characterize both the magnitude of the impulses and the influence of time delays on the function V . These effects are duly incorporated into the Lyapunov function, and the detailed proof is presented as follows.

The following Lyapunov function is proposed:

$$V(t) \triangleq V(\mathbf{e}(t)) = \mathbf{e}^T(t)(\mathbf{D} \otimes \mathbf{P})\mathbf{e}(t), \quad (12)$$

where $n \times n$ -matrix $\mathbf{P} > 0$. Evaluating $\dot{V}(t)$ along solutions of (6) yields:

$$\begin{aligned} \dot{V}(t) &= \dot{\mathbf{e}}^T(t)(\mathbf{D} \otimes \mathbf{P})\mathbf{e}(t) + \mathbf{e}^T(t)(\mathbf{D} \otimes \mathbf{P})\dot{\mathbf{e}}(t) \\ &= \mathbf{e}(t)^T[(\mathbf{I}_N \otimes \mathbf{A}) - (\mathbf{H} \otimes \mathbf{B}\mathbf{K})]^T(\mathbf{D} \otimes \mathbf{P})\mathbf{e}(t) \\ &\quad + \mathbf{e}^T(t)(\mathbf{D} \otimes \mathbf{P})[(\mathbf{I}_N \otimes \mathbf{A}) - (\mathbf{H} \otimes \mathbf{B}\mathbf{K})]\mathbf{e}(t) \\ &= \mathbf{e}^T(t)[\mathbf{D} \otimes \mathbf{A}^T \mathbf{P} - \mathbf{H}^T \mathbf{D} \otimes \mathbf{K}^T \mathbf{B}^T \mathbf{P}]\mathbf{e}(t) \\ &\quad + \mathbf{e}^T(t)[\mathbf{D} \otimes \mathbf{P} \mathbf{A} - \mathbf{D} \mathbf{H} \otimes \mathbf{P} \mathbf{B}\mathbf{K}]\mathbf{e}(t) \\ &= \mathbf{e}^T(t)[\mathbf{D} \otimes (\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A})]\mathbf{e}(t) \\ &\quad - \gamma \mathbf{e}^T(t)[(\mathbf{H}^T \mathbf{D} + \mathbf{D} \mathbf{H}) \otimes \mathbf{P} \mathbf{B} \mathbf{B}^T \mathbf{P}]\mathbf{e}(t) \\ &\leq \mathbf{e}^T(t)[\mathbf{D} \otimes (\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A})]\mathbf{e}(t) \\ &\quad - \mathbf{e}^T(t)[\mathbf{D} \otimes \rho_1 \mathbf{P} \mathbf{B} \mathbf{B}^T \mathbf{P}]\mathbf{e}(t) \end{aligned} \quad (13)$$

where $t \in [t_k, t_k + \varsigma_k)$, $k \in \mathbb{Z}_+$. By utilizing (3), it leads to

$$\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} - \rho_1 \mathbf{P} \mathbf{B} \mathbf{B}^T \mathbf{P} = -\rho_2 \mathbf{I}_n. \quad (14)$$

Further, by using (13)-(14), we can obtain that

$$\begin{aligned} \dot{V}(t) &\leq \mathbf{e}^T(t)[\mathbf{D} \otimes (\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} - \rho_1 \mathbf{P} \mathbf{B} \mathbf{B}^T \mathbf{P})]\mathbf{e}(t) \\ &= -\mathbf{e}^T(t)(\mathbf{D} \otimes \rho_2 \mathbf{I}_n)\mathbf{e}(t) \\ &\leq -\alpha_1 V(t), \end{aligned} \quad (15)$$

where $t \in [t_k, t_k + \varsigma_k)$, $k \in \mathbb{Z}_+$. From the result in (15), it further follows that

$$V(t) \leq V(t_k) e^{-\alpha_1(t-t_k)}, \quad t \in [t_k, t_k + \varsigma_k), \quad k \in \mathbb{Z}_+. \quad (16)$$

In addition, we can conclude from the first matrix inequality in (10) that

$$\begin{aligned}
 & V(t_k + \varsigma_k) \\
 &= \mathbf{e}^T(t_k + \varsigma_k)(\mathbf{D} \otimes \mathbf{P})\mathbf{e}(t_k + \varsigma_k) \\
 &\leq \mathbf{e}^T((t_k + \varsigma_k - \tau_{off})^-)(\mathbf{D} \otimes \mathbf{H}_k^T \mathbf{P} \mathbf{H}_k \mathbf{e})((t_k + \varsigma_k - \tau_{off})^-) \\
 &= \mathbf{e}^T((t_k + \varsigma_k - \tau_{off})^-)(\mathbf{D} \otimes \mathbf{H}_k^T \mathbf{P} \mathbf{P}^{-1} \mathbf{P} \mathbf{H}_k) \\
 &\quad \times \mathbf{e}((t_k + \varsigma_k - \tau_{off})^-) \\
 &< \mathbf{e}^T((t_k + \varsigma_k - \tau_{off})^-)(\mathbf{D} \otimes \beta_1 \mathbf{P})\mathbf{e}((t_k + \varsigma_k - \tau_{off})^-) \\
 &= \beta_1 V((t_k + \varsigma_k - \tau_{off})^-),
 \end{aligned} \tag{17}$$

for all $k \in \mathbb{Z}_+$. By referring to the derivation process of (13), we can derive from (9) that

$$\begin{aligned}
 \dot{V}(t) &= \dot{\mathbf{e}}^T(t)(\mathbf{D} \otimes \mathbf{P})\mathbf{e}(t) + \mathbf{e}^T(t)(\mathbf{D} \otimes \mathbf{P})\dot{\mathbf{e}}(t) \\
 &= \mathbf{e}(t)^T (\mathbf{I}_N \otimes \mathbf{A})^T (\mathbf{D} \otimes \mathbf{P})\mathbf{e}(t) \\
 &\quad + \mathbf{e}^T(t)(\mathbf{D} \otimes \mathbf{P}) (\mathbf{I}_N \otimes \mathbf{A}) \mathbf{e}(t) \\
 &< \alpha_2 V(t),
 \end{aligned}$$

for arbitrary $t \in (t_k + \varsigma_k, t_{k+1})$, $k \in \mathbb{Z}_+$, which results in

$$V(t) \leq V(t_k + \varsigma_k) e^{\alpha_2(t-t_k-\varsigma_k)}, \quad t \in [t_k + \varsigma_k, t_{k+1}). \tag{18}$$

Further, by using the second matrix inequality in (10), it yields that

$$\begin{aligned}
 V(t_{k+1}) &= \mathbf{e}^T(t_{k+1})(\mathbf{D} \otimes \mathbf{P})\mathbf{e}(t_{k+1}) \\
 &\leq \beta_2 V((t_{k+1} - \tau_{on})^-),
 \end{aligned} \tag{19}$$

for any $k \in \mathbb{Z}_+$. The subsequent discussions shall be carried out by (16)-(19).

For simplicity, let $\mathbf{e}(t) = \mathbf{e}(t, t_0, \mathbf{e}_0)$ stand for the solution to system (6) with initial value $(t_0, \mathbf{e}_0)$. It will be shown that

$$\begin{aligned}
 & V(t) \\
 &\leq \beta_1^k e^{k\alpha_1\tau_{off}} \beta_2^k e^{-k\alpha_2\tau_{on}} \exp \left[-(\alpha_1 + \alpha_2) \sum_{j=0}^{k-1} \varsigma_j \right] \\
 &\quad \times \exp[(\alpha_1 + \alpha_2)(t_k - t_0)] e^{-\alpha_1(t-t_0)} V_0, \quad t \in [t_k, t_k + \varsigma_k),
 \end{aligned} \tag{20}$$

and

$$\begin{aligned}
 & V(t) \\
 &\leq \beta_1^{k+1} e^{(k+1)\alpha_1\tau_{off}} \beta_2^k e^{-k\alpha_2\tau_{on}} \exp \left[-(\alpha_1 + \alpha_2) \sum_{j=0}^k \varsigma_j \right] \\
 &\quad \times \exp[(\alpha_1 + \alpha_2)(t - t_0)] e^{-\alpha_1(t-t_0)} V_0, \quad t \in [t_k + \varsigma_k, t_{k+1}),
 \end{aligned} \tag{21}$$

for any $k \in \mathbb{Z}_+$, where $V_0 = V(\mathbf{e}(t_0))$ and $\sum_{j=0}^{k-1} \varsigma_j \triangleq 0$ if $k = 0$. First, one can infer from (16) that

$$V(t) \leq e^{-\alpha_1(t-t_0)} V_0, \quad t \in [t_0, t_0 + \varsigma_0).$$

Subsequently, by using (17) and (18), it leads to

$$\begin{aligned}
 V(t) &\leq \beta_1 e^{-\alpha_1(t_0 + \varsigma_0 - \tau_{off} - t_0)} e^{\alpha_2(t-t_0-\varsigma_0)} V_0, \\
 &= \beta_1 e^{\alpha_1\tau_{off}} e^{-(\alpha_1 + \alpha_2)\varsigma_0} e^{(\alpha_1 + \alpha_2)(t-t_0)} e^{-\alpha_1(t-t_0)} V_0,
 \end{aligned}$$

for arbitrary $t \in [t_0 + \varsigma_0, t_1)$. Thus, (20) and (21) are true for $k = 0$. Next, suppose that (20) and (21) are also true for $k = N$, $N \geq 0$. That is,

$$\begin{aligned}
 & V(t) \\
 &\leq \beta_1^N e^{N\alpha_1\tau_{off}} \beta_2^N e^{-N\alpha_2\tau_{on}} \exp \left[-(\alpha_1 + \alpha_2) \sum_{j=0}^{N-1} \varsigma_j \right] \\
 &\quad \times \exp[(\alpha_1 + \alpha_2)(t_N - t_0)] e^{-\alpha_1(t-t_0)} V_0, \quad t \in [t_N, t_N + \varsigma_N),
 \end{aligned}$$

and for $t \in [t_N + \varsigma_N, t_{N+1})$

$$\begin{aligned}
 & V(t) \\
 &\leq \beta_1^{N+1} e^{(N+1)\alpha_1\tau_{off}} \beta_2^N e^{-N\alpha_2\tau_{on}} \\
 &\quad \times \exp \left[-(\alpha_1 + \alpha_2) \sum_{j=0}^N \varsigma_j \right] \\
 &\quad \times \exp[(\alpha_1 + \alpha_2)(t - t_0)] e^{-\alpha_1(t-t_0)} V_0.
 \end{aligned} \tag{22}$$

We will show that (20) and (21) are true when $k = N + 1$. Actually, it follows from (22), (19) and (16) that

$$\begin{aligned}
 & V(t) \\
 &\leq \beta_1^{N+1} e^{(N+1)\alpha_1\tau_{off}} \beta_2^{N+1} e^{-(N+1)\alpha_2\tau_{on}} \exp \left[-(\alpha_1 + \alpha_2) \right. \\
 &\quad \times \sum_{j=0}^N \varsigma_j \left. \right] \exp[(\alpha_1 + \alpha_2)(t_{N+1} - t_0)] e^{-\alpha_1(t_{N+1}-t_0)} \\
 &\quad \times e^{-\alpha_1(t-t_{N+1})} V_0, \\
 &= \beta_1^{N+1} e^{(N+1)\alpha_1\tau_{off}} \beta_2^{N+1} e^{-(N+1)\alpha_2\tau_{on}} \exp \left[-(\alpha_1 + \alpha_2) \right. \\
 &\quad \times \sum_{j=0}^N \varsigma_j \left. \right] \exp[(\alpha_1 + \alpha_2)(t_{N+1} - t_0)] e^{-\alpha_1(t-t_0)} V_0,
 \end{aligned} \tag{23}$$

for arbitrary $t \in [t_{N+1}, t_{N+1} + \varsigma_{N+1})$. Then, by using (23), (17) and (18), it leads to

$$\begin{aligned}
 & V(t) \\
 &\leq \beta_1^{N+2} e^{(N+2)\alpha_1\tau_{off}} \beta_2^{N+1} e^{-(N+1)\alpha_2\tau_{on}} \exp \left[-(\alpha_1 + \alpha_2) \right. \\
 &\quad \times \sum_{j=0}^N \varsigma_j \left. \right] \exp[(\alpha_1 + \alpha_2)(t_{N+1} - t_0)] e^{-\alpha_1(t_{N+1}+\varsigma_{N+1}-t_0)} \\
 &\quad \times e^{\alpha_2(t-t_{N+1}-\varsigma_{N+1})} V_0, \\
 &= \beta_1^{N+2} e^{(N+2)\alpha_1\tau_{off}} \beta_2^{N+1} e^{-(N+1)\alpha_2\tau_{on}} \exp \left[-(\alpha_1 + \alpha_2) \right. \\
 &\quad \times \sum_{j=0}^{N+1} \varsigma_j \left. \right] \exp[(\alpha_1 + \alpha_2)(t - t_0)] e^{-\alpha_1(t-t_0)} V_0,
 \end{aligned} \tag{24}$$

for arbitrary $t \in [t_{N+1} + \varsigma_{N+1}, t_{N+2})$.

Clearly, we can observe that (20) and (20) are fulfilled for $k = N + 1$. Consequently, the validity of (20) and (20) for any $k \in \mathbb{Z}_+$ is established via mathematical induction. Now, by

using the concepts of ADT and ACW on intermittent control sequences $\{t_k, \varsigma_k\}_{k \in \mathbb{Z}_+}$, it yields that

$$\begin{aligned} V(t) &\leq \max\{\beta_1, 1\} (\beta_1 e^{\alpha_1 \tau_{off}} \beta_2 e^{-\alpha_2 \tau_{on}})^{N(t, t_0)} \exp \left[-(\alpha_1 + \alpha_2) \right. \\ &\quad \times \left. \sum_{j=0}^{N(t, t_0)-1} \varsigma_j \right] e^{(\alpha_1 + \alpha_2)(t-t_0)} e^{-\alpha_1(t-t_0)} V_0 \\ &\leq \exp[N(t, t_0) \ln(\beta_1 e^{\alpha_1 \tau_{off}} \beta_2 e^{-\alpha_2 \tau_{on}})] \exp \left[-(\alpha_1 + \alpha_2) \right. \\ &\quad \times \left. \sum_{j=0}^{N(t, t_0)-1} \varsigma_j \right] e^{\alpha_2(t-t_0)} \max\{\beta_1, 1\} V_0 \end{aligned} \quad (25)$$

and

$$\begin{aligned} V(t) &\leq \exp \left\{ \left[\left(\frac{N(t, t_0) \ln(\beta_1 e^{\alpha_1 \tau_{off}} \beta_2 e^{-\alpha_2 \tau_{on}})}{N(t, t_0)} \right. \right. \right. \\ &\quad \left. \left. - \frac{(\alpha_1 + \alpha_2) \sum_{j=0}^{N(t, t_0)-1} \varsigma_j}{N(t, t_0)} \right) \div \frac{t-t_0}{N(t, t_0)} + \alpha_2 \right] \\ &\quad \times (t-t_0) \Big\} \max\{\beta_1, 1\} V_0 \end{aligned} \quad (26)$$

for all $t \geq t_0$, where $\max\{\beta_1, 1\}$ stands for the maximum between β_1 and 1, and $N(t, t_0)$ counts the control periods of intermittent control occurring within the interval $(t_0, t]$. By using conditions (7) and (8), one can further obtain from (11) that

$$\begin{aligned} \lim_{t \rightarrow +\infty} &\left[\left(\frac{N(t, t_0) \ln(\beta_1 e^{\alpha_1 \tau_{off}} \beta_2 e^{-\alpha_2 \tau_{on}})}{N(t, t_0)} \right. \right. \\ &\quad \left. \left. - \frac{(\alpha_1 + \alpha_2) \sum_{j=0}^{N(t, t_0)-1} \varsigma_j}{N(t, t_0)} \right) \div \frac{t-t_0}{N(t, t_0)} + \alpha_2 \right] \\ &= \frac{\ln \beta_1 + \ln \beta_2 + \alpha_1(\tau_{off} - \bar{\varsigma}) + \alpha_2(T_a - \bar{\varsigma} - \tau_{on})}{T_a} \\ &< 0. \end{aligned} \quad (27)$$

Clearly, we can conclude from (26) and (27) that $V(t) \rightarrow 0$ as $t \rightarrow +\infty$. Together with (12), this proves that the average-width intermittent controller $(\mathbf{K}, \{t_k\}, \{\varsigma_k\})$ ensures leader-following consensus in systems (1) and (2). This concludes the proof. $\square$

Remark 5. In most existing results on AIC, the control widths are typically constrained by fixed lower or upper bounds. Meanwhile, the “on-off” switching behavior inherent in intermittent control may induce impulsive effects, whose presence often imposes additional restrictions such as upper bounds on control widths or impulsive intervals ([25–27]). These requirements may lead to inflexible consensus or stability criteria. In contrast, this study relies only on the average values

(i.e., ADT and ACW), rather than requiring each individual control period or width to satisfy certain bounds. Moreover, both the ADT and ACW conditions are adopted in a limiting form, which together constitute a more flexible framework. Consequently, the sufficient conditions derived in Theorem 1 accommodate both intermittent control and impulsive effects under milder constraints. The current work mainly focuses on linear models, while extensions to nonlinear cases are left for future research.

Remark 6. Condition (11) essentially requires that, over an average control cycle, the overall stabilizing effect must outweigh the overall destabilizing effect. The parameters β_1 and β_2 quantify the amplification/attenuation of the Lyapunov function at the “off” and “on” switching instants, respectively. When $\mathbf{H}_k$ (or $\mathbf{G}_k$) acts overall as an expanding mapping, the corresponding β_1 (or β_2) is typically greater than 1; conversely, if it acts as a contracting mapping, the corresponding β can be less than 1. Specifically, a value of $\beta > 1$ indicates that the impulse is disruptive, while $\beta < 1$ indicates that it is facilitating. Accordingly, this term reflects the net contribution of the impulse strengths. The term $\alpha_1(\tau_{off} - \bar{\varsigma})$ reflects the combined effect at the “off” instant: $-\alpha_1 \bar{\varsigma}$ represents the contribution of the average control phase (stabilizing), while $+\alpha_1 \tau_{off}$ represents the contribution of the impulse delay at the “off” instant (typically disruptive). This shows that a larger $\bar{\varsigma}$ helps counteract the adverse effect of τ_{off} .

Referring to the previous results [25–27], we further consider the specific case that there exists no delay in impulses, that is, $\tau_{off} = \tau_{on} = 0$.

Corollary 1. Under Assumptions 1-2, suppose that given constants $\alpha_2 > 0$, $\beta_1 > 0$ and $\beta_2 > 0$, the LMIs (9)-(10) are satisfied. Then, the average-width intermittent controller $(\mathbf{K}, \{t_k\}, \{\varsigma_k\})$ ensures leader-following consensus in systems (1) and (2) with delayed impulses (5), if following condition is satisfied:

$$\ln \beta_1 + \ln \beta_2 - \alpha_1 \bar{\varsigma} + \alpha_2(T_a - \bar{\varsigma}) < 0. \quad (28)$$

Remark 7. Actually, one can see from Theorem 1 with $\alpha_1 > 0$ that condition (11) turns to be easier to hold if increasing delay τ_{on} or decreasing delay τ_{off} . Compared with Theorem 1 and Corollary 1, we can further reveal the effects of delays τ_{off} and τ_{on} on the consensus of the considered multi-agent systems. To be specific, we can conclude from (10) that the existence of delay τ_{off} may destroy the consensus but delay τ_{on} may be beneficial for the consensus. Namely, original consensus systems (1) and (2) may become inconsistent by increasing delay τ_{off} , conversely, original inconsistent systems (1) and (2) may turn to be consensus by increasing delay τ_{on} . An example illustrating these results can be found in Section V.

Furthermore, if there exists no impulses on intermittent control, that is, $\mathbf{H}_k = \mathbf{G}_k = \mathbf{I}_n$ for all $k \in \mathbb{Z}_+$ and $\tau_{off} = \tau_{on} = 0$. Then, it leads to $\beta_1 = \beta_2 = 1$, which yields the following result.

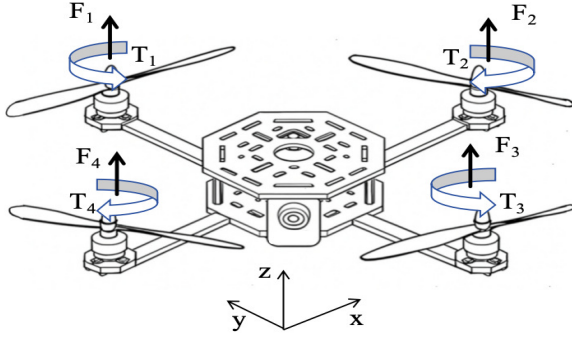


Fig. 1. Structure diagram of a quadrotor UAV.

Corollary 2. Under Assumptions 1-2, suppose that given constant $\alpha_2 > 0$, (9) is satisfied. Then, the average-width intermittent controller $(\mathbf{K}, \{t_k\}, \{\varsigma_k\})$ ensures leader-following consensus in systems (1) and (2) if following condition is satisfied:

$$-\alpha_1 \bar{\varsigma} + \alpha_2 (T_a - \bar{\varsigma}) < 0. \quad (29)$$

Remark 8. From both Theorem 1 and Corollary 2, we can observe that the larger ACW constant $\bar{\varsigma}$ or the smaller ADT constant T_a is conducive to the satisfaction of condition (11) and condition (28). It implies that two originally inconsistent multi-agent systems under intermittent control can achieve consensus by increasing the ACW or decreasing the ADT. In addition, it follows from Theorem 1 that condition (11) may no longer be satisfied if coefficients β_1 and β_2 become larger. Namely, originally consistent systems (1) and (2) under AIC (4) will be inconsistent if there exist the consensus-disruptive impulses (when $\beta_1 > 1$, $\beta_2 > 1$), or conversely, originally inconsistent systems (1) and (2) under AIC (4) may be consensus if there exist the consensus-promoting impulses (when $\beta_1 < 1$, $\beta_2 < 1$). Some detailed discussions with an array of numerical simulations can be seen in Section V.

IV. APPLICATION

In this section, we shall apply the derived results to the flexible intermittent control of Quadrotor UAVs. It is known that a quadrotor UAV exhibits six independent degrees of freedom in relation to an inertial reference frame, encompassing linear displacements along the X , Y , and Z axes, as well as rotational movements characterized by the Euler angles φ (indicating roll), θ (representing pitch), and ψ (corresponding to yaw). The control of quadrotors is achieved by adjusting the thrust generated by each rotor, which is denoted as F_i in Fig. 1. The state vector can be expressed as

$$\mathbf{x} = [\dot{\varphi} \ \dot{\theta} \ \dot{\psi} \ \varphi \ \theta \ \psi \ \dot{X} \ \dot{Y} \ \dot{Z} \ X \ Y \ Z]^T.$$

The quadrotor's dynamic behavior is characterized by the

following nonlinear equations [47]:

$$\begin{cases} \ddot{\phi} = \frac{I_{yy} - I_{zz}}{I_{xx}} \dot{\theta} \dot{\psi} - \frac{J_{TP}}{I_{xx}} \dot{\theta} \Omega + \frac{u_2}{I_{xx}} \\ \ddot{\theta} = \frac{I_{zz} - I_{xx}}{I_{yy}} \dot{\phi} \dot{\psi} + \frac{J_{TP}}{I_{yy}} \dot{\phi} \Omega + \frac{u_3}{I_{yy}} \\ \ddot{\psi} = \frac{I_{xx} - I_{yy}}{I_{zz}} \dot{\phi} \dot{\theta} + \frac{u_4}{I_{zz}} \\ \ddot{X} = (\sin(\psi) \sin(\phi) + \cos(\psi) \sin(\theta) \cos(\phi)) \frac{u_1}{M} \\ \ddot{Y} = (-\cos(\psi) \sin(\phi) + \sin(\psi) \sin(\theta) \cos(\phi)) \frac{u_1}{M} \\ \ddot{Z} = -g + (\cos(\theta) \cos(\phi)) \frac{u_1}{M} \end{cases} \quad (30)$$

where M is the vehicle's mass, g is the gravitational acceleration near Earth's surface, I_{kk} denotes the moment of inertia about each body axis k , J_{TP} represents the propeller's rotational inertia around the propeller axis, and $\Omega = -\Omega_1 + \Omega_2 - \Omega_3 + \Omega_4$, with $\Omega_j, j \in \{1, 2, 3, 4\}$ being each propeller's individual speed. The parameter J_{TP} is substantially smaller than the body's moments of inertia and is consequently disregarded in our analysis.

Since the quadrotor predominantly operates in hovering conditions, it is reasonable to set the attitude angles, their rates, and the vehicle velocity to zero at the operating point. For this investigation, we utilize the model (30) after its linearization about the zero equilibrium state. This particular state corresponds to the vehicle's hovering maneuver, with the resulting system specified below [48]:

$$\dot{\mathbf{x}}_i(t) = \mathbf{A}_q \mathbf{x}_i(t) + \mathbf{B}_q \mathbf{u}_i(t), \quad i = 1, \dots, N, \quad t \geq t_0 \quad (31)$$

and

$$\dot{\mathbf{x}}_0(t) = \mathbf{A}_q \mathbf{x}_0(t), \quad t \geq t_0 \geq 0, \quad (32)$$

where $\mathbf{x}_i, \mathbf{x}_0 \in \mathbb{R}^{12}$ and $\mathbf{u}_i \in \mathbb{R}^4$ represent the states of the followers quadrotors, the state of the leader quadrotor, and the control input, respectively, and

$$\mathbf{A}_q = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & g & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -g & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \quad (33)$$

$$\mathbf{B}_q = \begin{bmatrix} 0 & \frac{1}{I_{xx}} & 0 & 0 \\ 0 & 0 & \frac{1}{I_{yy}} & 0 \\ 0 & 0 & 0 & \frac{1}{I_{zz}} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{M} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (34)$$

In practical quadrotor UAVs networks, continuous communication among agents is often infeasible or undesirable due to the need to conserve energy and communication bandwidth [49]. Intermittent control, by its very nature, is a strategy designed to address this challenge by saving these critical resources. Consider AIC $u_i(t)$ described by (4). Motivated by recent work on intermittent control with impulses [25–27, 49], we further account for the influence of delays on the impulse. Namely, we consider quadrotor UAVs (31)-(32) with delayed impulse (5).

This linear model has motivated a variety of interesting results in the literature [48, 50, 51]. However, these studies have predominantly focused on the stability analysis or tracking problem for a single UAV. In real-world scenarios, collaborative operation among multiple UAVs is often required [52, 53]. Therefore, this paper investigates the leader-following consensus problem for a system of multiple quadrotors. In particular, by applying Theorem 1, the following theorem holds:

Theorem 2. *Under Assumptions 1 and 2, consider the linearized model of the quadrotor (31)-(32) under intermittent controller (4) with delayed impulse (5). Assume that there exist constants $\alpha_2 > 0$, $\beta_1 > 0$ and $\beta_2 > 0$ satisfying (10), (11), the following inequality hold:*

$$\mathbf{A}_q^T \mathbf{P} + \mathbf{P} \mathbf{A}_q - \alpha_2 \mathbf{P} < 0 \quad (35)$$

Then, leader-following consensus of model (31)-(32) can be achieved via average-width intermittent controller $(\mathbf{K}, \{t_k\}, \{\varsigma_k\})$, where $\mathbf{K} = \gamma \mathbf{B}_q^T \mathbf{P}$.

Proof. Based on a methodology comparable to that of Theorem 1, the proof is not repeated here. $\square$

Remark 9. First, this paper investigates the consensus problem for a multi-UAV system, shifting the focus from single-agent control [48, 50, 51] to cooperative networked control. Second, regarding the control strategy, while some literature has explored intermittent control for UAVs (e.g., [54]), such methods often impose a constraint on the upper bound of the control width. The approach developed in this study is less restrictive as it removes this condition, thus offering greater flexibility in control design. Finally, the proposed framework further considers the potential effects of delayed impulses, which are inevitable in practical scenarios. In particular, the

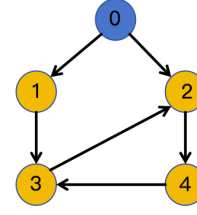


Fig. 2. Communication graph $\mathcal{G}_1$ of Example 1.

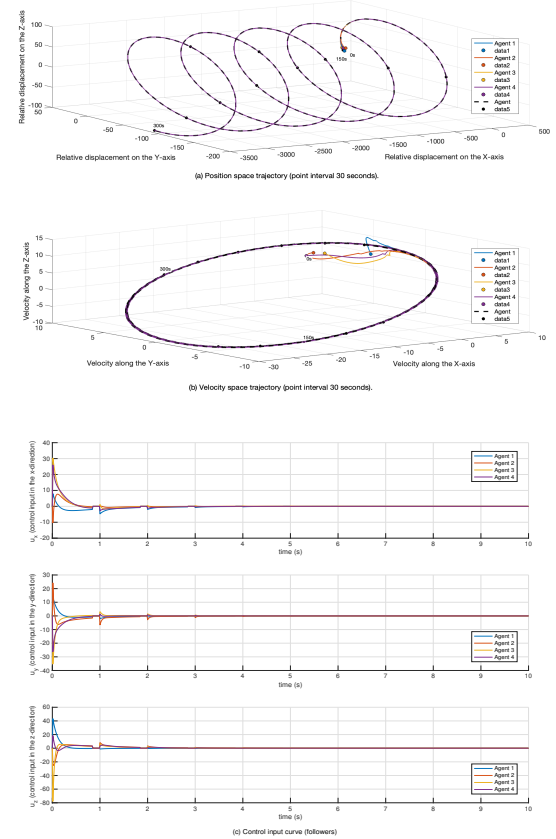


Fig. 3. The trajectories and control inputs of the agents under average-width intermittent control with delay-free impulses (where $\tau_{on} = 0$, $\tau_{off} = 0$).

simulation results of the UAV application case correspond to the second example in Section V.

V. ILLUSTRATIVE EXAMPLES

To demonstrate the efficacy of the theoretical framework developed in prior sections, two simulated cases are examined.

Example 1. In this example, each agent is a spacecraft in low Earth orbit, forming a system with six spacecraft, as investigated in [55]. After transformation, the spacecraft formation issue can be turned into a consensus problem for a linear multi-agent system. The dynamics of each agent is

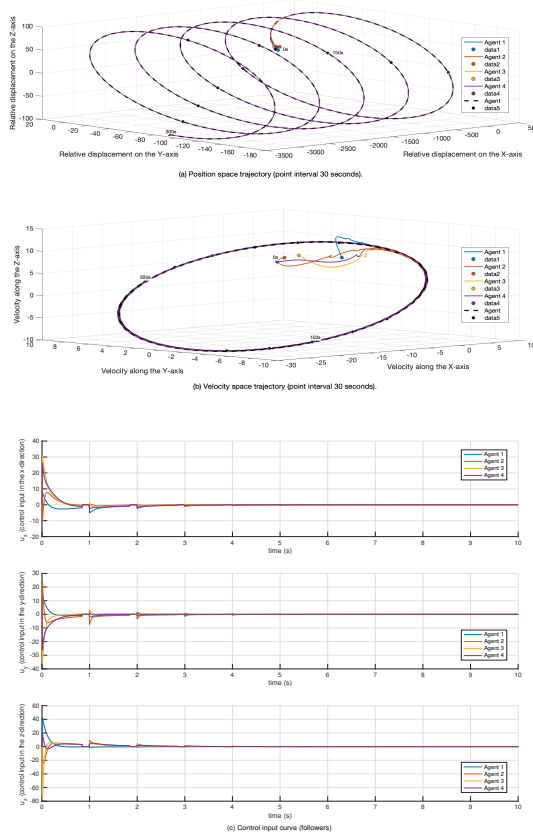


Fig. 4. The trajectories and control inputs of the agents under the average-width intermittent control with delayed impulses (where $\tau_{on} = 0$, $\tau_{off} = 0.11$).

modeled as follows:

$$\mathbf{A} = \begin{bmatrix} 0_3 & \mathbf{I}_3 \\ \mathbf{A}_1 & \mathbf{A}_2 \end{bmatrix}, \quad \mathbf{A}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3\omega^2 & 0 \\ 0 & 0 & -\omega^2 \end{bmatrix},$$

$$\mathbf{A}_2 = \begin{bmatrix} 0 & 2\omega & 0 \\ -2\omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0_3 \\ \mathbf{I}_3 \end{bmatrix},$$

where ω represents the satellite's angular rate and $\omega = 0.1$. In particular, the state is represented as $\mathbf{x}_i = \text{col}(\bar{x}_i, \bar{y}_i, \bar{z}_i, v_x^i, v_y^i, v_z^i)$, where $(\bar{x}_i, \bar{y}_i, \bar{z}_i)$ denotes the displacement from the target position in the X-Y-Z axes, and (v_x^i, v_y^i, v_z^i) indicates the velocity along these axes.

Clearly, Assumption 1 holds. The communication topology among agents is depicted in Fig. 2. Furthermore, Assumption 2 is confirmed. The matrix $\mathbf{H}$ is defined as: $[1, 0, 0, 0; 0, 2, -1, 0; -1, 0, 2, -1; 0, -1, 0, 1]$. Setting $\mathbf{D} = \mathbf{I}_4$, it is easy to verify that $\mathbf{DH} + \mathbf{H}^T\mathbf{D}$ is positive definite. It can be obtained by solving the Matlab matrix function that $\gamma = 12.1260$, $\mathbf{P} = [5.6647, 0.0003, 0, 0.7069, 0.0177, 0; 0.0003, 5.6674, 0, -0.0177, 0.7099, 0; 0, 0, 5.6623, 0, 0, 0.7061; 0.7069, -0.0177, 0, 0.8009, -0.0000, 0; 0.0177, 0.7099, 0, -0.0000, 0.8012, 0; 0, 0, 0.7061, 0, 0, 0.8008]$. The gain matrix $\mathbf{K}$ for the intermittent controller (4) is $[8.5717, -0.2145, 0.97113, -0.0001, 0; 0.2145, 8.6082, 0, -0.0001, 9.7158, 0; 0, 0, 8.5623, 0, 0, 9.7101]$.

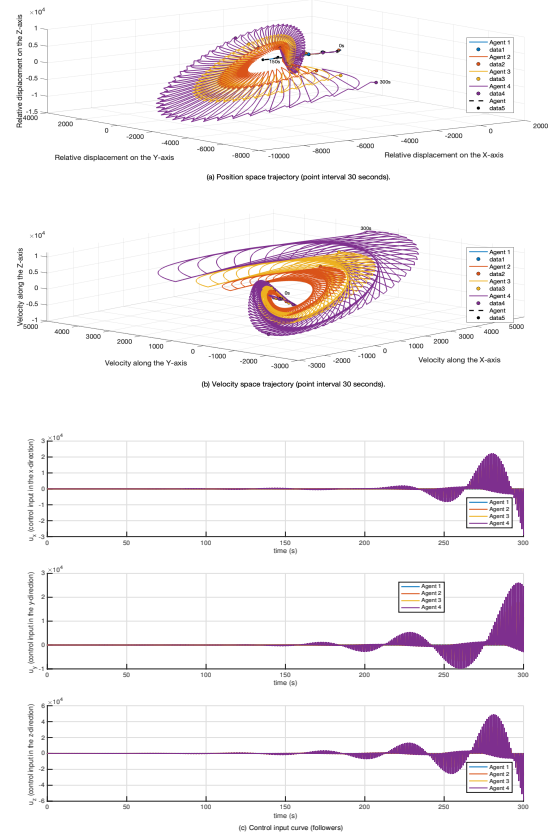


Fig. 5. The trajectories and control inputs of the agents under the average-width intermittent control with delayed impulses (where $\tau_{on} = 0$, $\tau_{off} = 0.84$).

Consider impulsive gain matrices $\mathbf{H}_k = \mathbf{G}_k = 1.04\mathbf{I}_6$ and delays in impulses $\tau_{off} = \tau_{on} = 0$. Design control sequence $\{t_k, \varsigma_k\}_{k \in \mathbb{Z}_+}$ satisfying $t_k = k$, $\varsigma_k = 0.85$, where $k \in \mathbb{Z}_+$, which leads to $T_a = 1$ and $\bar{\varsigma} = 0.85$. Then, one can verify that $\ln \beta_1 + \ln \beta_2 - \alpha_1 \bar{\varsigma} + \alpha_2 (T_a - \bar{\varsigma}) < 0$, with the parameters $\alpha_1 = 0.8667$, $\alpha_2 = 3$, and $\beta_1 = \beta_2 = 1.1$ satisfying all the conditions in Corollary 1. Then, we can conclude that system (1) and (2) can achieve leader-following consensus under average-width intermittent control (4) with delay-free impulses (see Fig. 3 for the illustrations of trajectories and control inputs). Further, when $\tau_{on} = 0$ and $\tau_{off} < \frac{-\ln \beta_1 + \ln \beta_2 - \alpha_1 \bar{\varsigma} + \alpha_2 (T_a - \bar{\varsigma})}{\alpha_1} = 0.11088$, the system can achieve consensus, as shown in Fig. 4 (where $\tau_{on} = 0$, $\tau_{off} = 0.11$). If the time delays exceed this range, the consensus may not be maintained, which is illustrated in Fig. 5 (where $\tau_{on} = 0$, $\tau_{off} = 0.84$). It implies that the delay τ_{off} can induce consensus-disruptive effects on the consensus of systems (1) and (2) under average-width intermittent control (4) with delayed impulses at “off” switching instants.

Next, we analyze how the delay τ_{on} facilitates consensus in the impulses at “on” switching instants. Consider impulsive gain matrices $\mathbf{H}_k = \mathbf{G}_k = 0.9\mathbf{I}_6$ and delays in impulses $\tau_{off} = \tau_{on} = 0$. We analyze multi-agent system (1) and (2) under average-width intermittent control (4), where control sequence $\{t_k, \varsigma_k\}_{k \in \mathbb{Z}_+}$ satisfies $t_k = 10k$, $\varsigma_k = 0.2$ for all

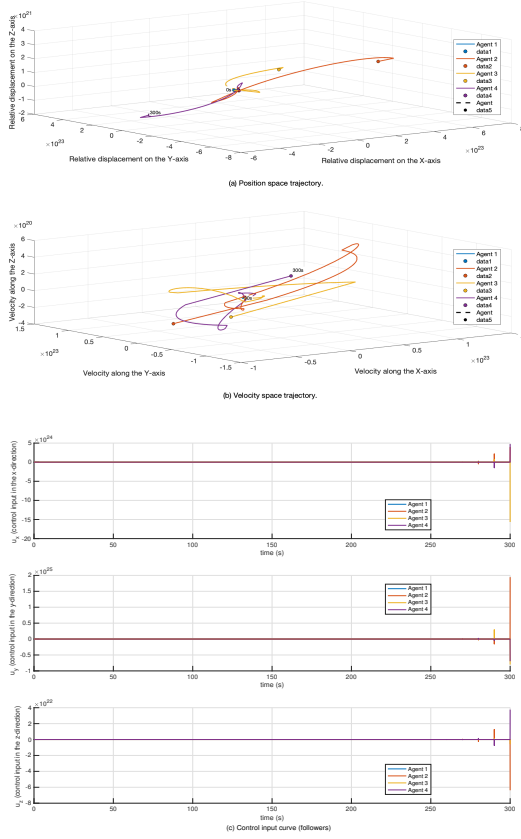


Fig. 6. The trajectories and control inputs of the agents under the average-width intermittent control with delay-free impulses (where $\tau_{on} = 0$, $\tau_{off} = 0$).

$k \in \mathbb{Z}_+$, which leads to $T_a = 10$ and $\bar{\zeta} = 0.2$. In particular, other conditions and assumptions are the same as those in the above case. Then, we can check that $\ln \beta_1 + \ln \beta_2 - \alpha_1 \bar{\zeta} + \alpha_2(T_a - \bar{\zeta}) \not\leq 0$, for given $\alpha_1 = 0.3337$, $\alpha_2 = 0.66$ and $\beta_1 = \beta_2 = 0.9$. That is to say the condition (28) in Corollary 1 is not fulfilled, which implies that multi-agent system (1) and (2) under average-width intermittent control (4) with delay-free impulses may fail to achieve consensus (see Fig. 6 for the illustrations of trajectories and control inputs). When $\tau_{off} = 0$ and $\tau_{on} > \frac{\ln \beta_1 + \ln \beta_2 - \alpha_1 \bar{\zeta} + \alpha_2(T_a - \bar{\zeta})}{\alpha_2} \approx 9.67197$, the system can achieve consensus, as shown in Fig. 7 (where $\tau_{on} = 9.68$, $\tau_{off} = 0$) and Fig. 8 (where $\tau_{on} = 9.99$, $\tau_{off} = 0$). It implies that, under average-width intermittent control (4) with delayed impulses at “off” switching instants, the delay τ_{on} can exert consensus-promoting effects on systems (1) and (2). As shown in (11), increasing τ_{on} directly enhances the negativity of $-\alpha_2 \tau_{on}$, thereby strengthening the overall attenuation effect. This facilitates a net decrease in energy over each cycle and ultimately leads to consensus.

In contrast to previous research, which often attributes to time delay a monolithic effect that is either beneficial or detrimental (for example, [32, 33] considered separately the impact of different types of time delays on system consensus or stability, whereas [15, 31, 34, 42] focused on their comprehensive or average effects), a key contribution of this

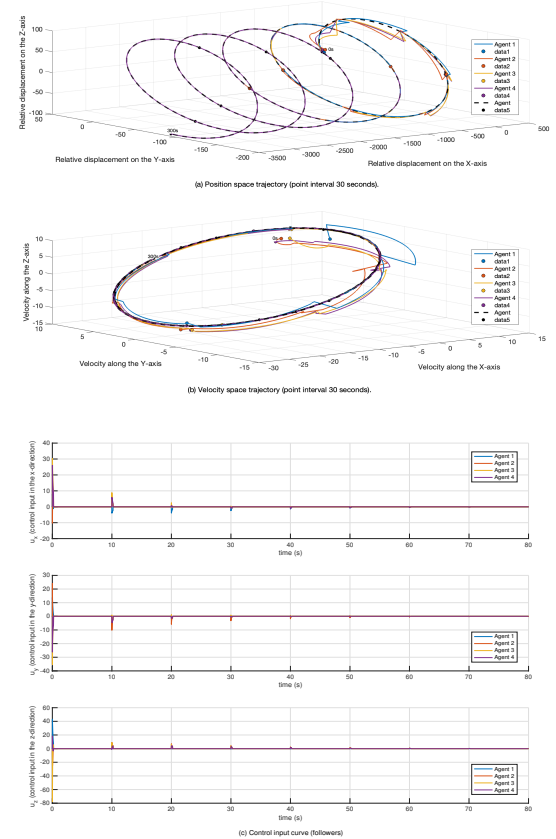


Fig. 7. The trajectories and control inputs of the agents under the average-width intermittent control with delay-free impulses (where $\tau_{on} = 9.68$, $\tau_{off} = 0$).

TABLE I
DELAY PARAMETERS

Item	1	2	3	4	5	6
τ_{on}	0	0	0	0	9.68	9.99
τ_{off}	0	0.11	0.84	0	0	0
LHS of Eq. (11)	-0.0961	-0.000763	0.631928	29.0159	-0.0241	-0.9541
consensus	yes	yes	no	no	yes	yes
state trajectories	Fig. 3	Fig. 4	Fig. 5	Fig. 6	Fig. 7	Fig. 8

paper lies in the fine-grained differentiation and decoupling of different types of impulsive delays. Through our analytical framework, the ‘on’-instant delay (τ_{on}) and the ‘off’-instant delay (τ_{off}) are treated as independent parameters, thereby clearly revealing their distinct mechanisms of action on system consensus. Specifically, both theoretical analysis and numerical simulations demonstrate that τ_{on} can play a constructive role as a “promoting term” under certain conditions, facilitating the achievement of consensus. Conversely, τ_{off} acts as an “inhibiting term”, whose presence adversely affects the consensus process. The variations of the delay parameters are presented in Table I.

Example 2. Consider a multi-agent system consisting of one leader quadrotor and five follower quadrotors. For the prototype depicted in Fig. 1, with parameters $I_{xx} = I_{yy} = 3.1 \times 10^{-3} \text{N} \cdot \text{m} \cdot \text{s}^2$, $I_{zz} = 25.5 \times 10^{-3} \text{N} \cdot \text{m} \cdot \text{s}^2$, and $M = 0.987 \text{kg}$ (see [48, 50]), the matrix parameters of

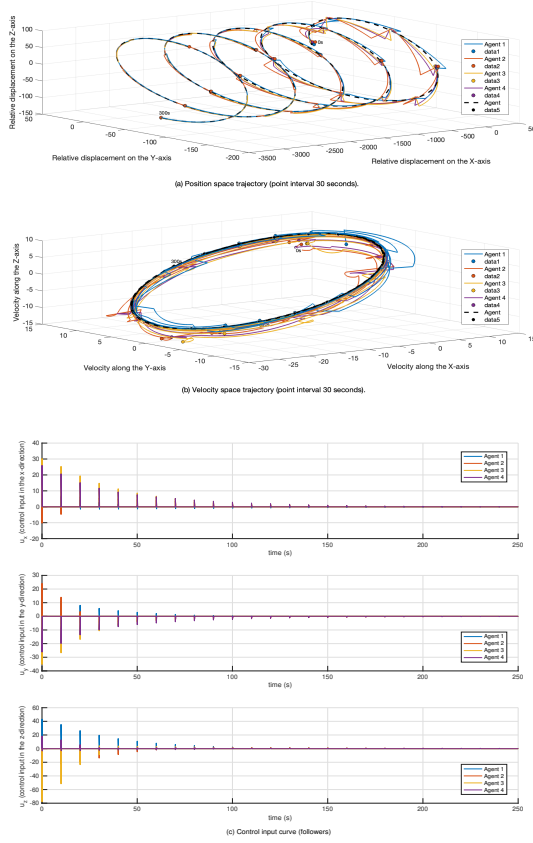


Fig. 8. The trajectories and control inputs of the agents under the average-width intermittent control with delay-free impulses (where $\tau_{on} = 9.99$, $\tau_{off} = 0$).

linearized quadrotor model (31) and (32) can be derived as follows:

$$\mathbf{A}_q = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 9.81 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -9.81 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix},$$

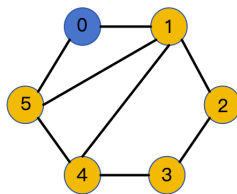


Fig. 9. Communication graph $\mathcal{G}_1$ of Example 2.

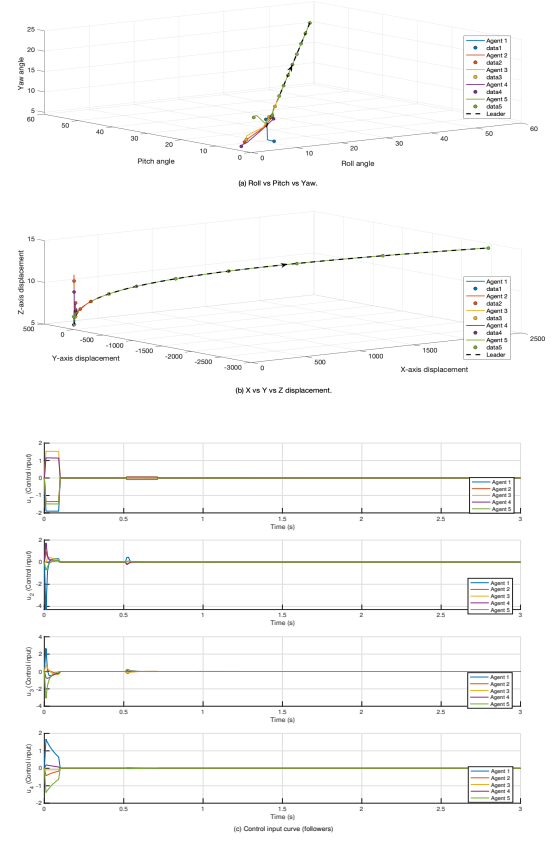


Fig. 10. The trajectories and control inputs of the agents under the average-width intermittent control in Example 2.

$$\mathbf{B}_q = \begin{bmatrix} 0 & 322.5806 & 0 & 0 \\ 0 & 0 & 322.5806 & 0 \\ 0 & 0 & 0 & 39.2157 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1.0132 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Evidently, Assumption 1 is met. The communication topology among agents is illustrated in Fig. 9. Additionally, Assumption 2 is verified. $\mathbf{H} = [4, -1, 0, -1, -1; -1, 2, -1, 0, 0; 0, -1, 2, -1, 0; -1, 0, -1, 3, -1; -1, 0, 0, -1, 3]$. Set $\mathbf{D} = \mathbf{I}_5$. It is straightforward to confirm the positive definiteness of $\mathbf{DH} + \mathbf{H}^T \mathbf{D}$. It can be obtained by solving the Matlab matrix function that $\gamma = 0.17269$, $\mathbf{P} = [0.0010, 0, 0, 0.0055, 0, 0, 0, -0.0014, 0, 0, -0.0010, 0; 0, 0.0010, 0, 0, 0.0055, 0, 0.0014, 0, 0, 0.0010, 0, 0; 0, 0, 0.0087, 0, 0, 0.0081, 0, 0, 0, 0, 0, 0; 0.0055, 0, 0, 0.5723, 0, 0, 0, -0.1529, 0, 0, -0.1053, 0; 0, 0.0055, 0, 0, 0.5723, 0, 0.1529, 0, 0, 0.1053, 0, 0; 0, 0, 0.0081, 0, 0, 0.1078, 0, 0, 0, 0, 0, 0; 0, 0, 0.0014, 0, 0, 0.1529,$

0, 0.0722, 0, 0, 0.0568, 0, 0; -0.0014, 0, 0, -0.1529, 0, 0, 0, 0.0722, 0, 0, 0.0568, 0; 0, 0, 0, 0, 0, 0, 0, 0.8399, 0, 0, 0.3121; 0, 0.0010, 0, 0, 0.1053, 0, 0.0568, 0, 0, 0.1461, 0, 0; -0.0010, 0, 0, -0.1053, 0, 0, 0.0568, 0, 0, 0.1461, 0; 0, 0, 0, 0, 0, 0, 0.3121, 0, 0, 0.2691]. The control gain $\mathbf{K}$ for the intermittent control law (4) is $[0, 0, 0, 0, 0, 0, 0, 0, 0.1470, 0, 0, 0.0546; 0.05751, 0, 0, 0.30407, 0, 0, 0, -0.07980, 0, 0, -0.05461, 0; 0, 0.05751, 0, 0, 0.30407, 0, 0.07980, 0, 0, 0.05461, 0, 0; 0, 0, 0.05885, 0, 0, 0.05461, 0, 0, 0, 0, 0, 0]$. When intermittent control sequence $\{t_k, \varsigma_k\}_{k \in \mathbb{Z}_+}$ is given by $t_0 = 0, t_k = 0.5k + 0.02 \times (-1)^{k+1}, k = 1, 2, \dots$ and $\varsigma_0 = 0.1, \varsigma_1 = 0.2, \varsigma_2 = 0.3, \varsigma_3 = 0.2, \varsigma_j = 0.4 + 0.02 \times (-1)^{j+1}, j = 4, 5, \dots$, then it leads to $T_a = 0.5, \bar{\varsigma} = 0.4$. If there emerge impulses with gain matrices $\mathbf{H}_k = \mathbf{G}_k = 0.2\mathbf{I}_{12}$, then one can check that $\ln \beta_1 + \ln \beta_2 - (\alpha_1 + \alpha_2)\bar{\varsigma} + \alpha_2 T_a < 0$, and all the conditions in Theorem 2 are fulfilled for given $\alpha_1 = 0.1023, \alpha_2 = 27$ and $\beta_1 = \beta_2 = 0.05$. Then, we can conclude that multi-agent system (31) and (32) can achieve leader-following consensus under average-width intermittent control (4) with impulses, which matches the simulations in Fig. 10.

Although the concept of average control rate was given in [13], the concept of ACW proposed in this paper contains more intuitive sense and is easier to verify in some situations. Furthermore, studies such as [4, 13, 56] have not considered the impulsive effects triggered by “on-off” switching behaviors, whereas the model in this paper explicitly incorporates such impulses with time delays. In addition, compared with the control strategies adopting periodic or impulse-free schemes in [4, 7, 8], the method proposed in this paper integrates aperiodic control and delayed impulses within a unified framework, thus providing a more flexible solution that is closer to practical scenarios. Furthermore, unlike previous single-agent control studies [48, 50, 51], this paper investigates multi-UAV collaborative consensus and, compared to the bounded control width in [54], proposes an ACW approach that eliminates this constraint for greater flexibility. All simulations are conducted using MATLAB tools. Based on the MATLAB differential equation solver and matrix operation toolbox, we construct the dynamic model, the impulsive control algorithm, and the time-delay simulation module, and complete the state trajectory analysis and data calculation under multiple scenarios.

VI. CONCLUSION

The proposed average-width intermittent control framework provides a flexible and efficient solution for cooperative control of multi-agent systems in scenarios with limited communication and computational resources. In this study, the concept of ACW in limiting form is introduced to globally characterize the control widths, and flexible consensus criteria are derived based on ACW and ADT. In particular, the delayed impulses induced by “on-off” switching behaviors are investigated, revealing that delays at “off” instants are consensus-disruptive, while those at “on” instants are consensus-facilitating. The effectiveness of the proposed scheme is demonstrated through a leader-following task for quadrotor UAVs.

The theoretical significance of this framework lies in its ability to differentiate the effects of impulse delays and to relax

strict constraints on control timing via average-type conditions, making it well-suited for energy-constrained and non-ideal network scenarios such as multi-UAV formation coordination and collaborative monitoring in sensor networks. However, its effective application relies on assumptions including communication topology connectivity and linear system models. Namely, the theoretical analysis itself is developed within the linear framework, which implies that the results cannot be directly applied to nonlinear models. Actually, we can investigate local consensus by using linearization techniques, or handle the nonlinear terms under the Lipschitz condition (see [4, 5]). In particular, systems with strong nonlinearities, actuator saturation, or more complex network uncertainties remain important directions for future research.

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